BITAINTEL

B2B Business Verification, Intelligence & Trust Platform

"Verify Before You Trust"

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Phase 1 — Scope, Technical Architecture, Effort Estimation, Cost & Timeline

Prepared by: Softfix
Lucknow, India
Document Date: September 2026 | Version 1.0

1. Introduction & Objective

BITAINTEL is proposed as a B2B business verification and intelligence platform that helps companies identify, verify, evaluate, compare, and connect with reliable business partners, manufacturers, suppliers, distributors, and customers.

The objective of Phase 1 is to establish a trusted B2B business identity and verification ecosystem, built around structured data collection, government-backed verification, a transparent scoring engine, and a searchable directory of verified businesses.

1.1 Core Principle

"Verify Before You Trust." BITAINTEL is not positioned as a simple business directory; it is a structured business trust & intelligence profile, differentiated from directory/marketplace competitors through verification, scoring, and intelligence.

1.2 Purpose of This Document

This SRS defines the functional scope, technical architecture, database structure, user roles, security requirements, effort estimation, cost, and delivery timeline for Phase 1 development, using Laravel as the core backend framework as requested.

2. Project Scope

2.1 Phase 1 — In Scope (Must-Have for Launch)

- BITAINTEL landing website
- Business/vendor registration
- Secure login (email + mobile OTP)
- Email/mobile verification
- Business profile creation (multi-module)
- GSTIN verification (via GSP/ASP integration)
- PAN / CIN / LLPIN identity capture
- Company document upload & versioning
- Verification workflow engine
- Manual verification by BITAINTEL team
- Multi-level verification status
- BITAINTEL verification badge
- Business scoring engine (weighted, 0-100)
- Vendor/business search with filters

- Public business profile page
- Business comparison (2-4 businesses)
- Vendor dashboard
- Admin dashboard
- Verification team dashboard
- Document review workflow
- Approval / rejection / clarification workflow
- Re-verification workflow
- Notification system (email, SMS/OTP)
- Complaint/dispute management
- Audit logs (full trail)
- User / role management (RBAC — 6 roles)
- Reports & analytics
- API integration layer (pluggable government/API providers)
- Data security & access control
- Privacy / consent management
- Responsive web application

2.2 Phase 2 — Explicitly Out of Scope (Future Roadmap)

The following are kept outside Phase 1 unless development capacity and budget are separately expanded:

- AI-powered company analysis & AI-generated risk report
- AI document intelligence / inconsistency detection
- Automated financial analysis & credit-risk assessment
- Global/international company verification & sanctions screening
- Export/import data analysis, manufacturer capability & supply-chain intelligence
- Automated recommendation / AI-powered supplier matching engine
- Public API access for external companies
- Enterprise subscription & paid verification packages
- Advanced BI reports
- Mobile application (iOS/Android)
- WhatsApp integration
- Automated voice verification

- Blockchain-based verification certificate
- International GST/VAT & global regulatory database integrations

Phase 2 items are excluded from the cost and timeline estimates in this document and will be scoped separately once Phase 1 is live and validated.

3. Technology Stack (Laravel-based)

3.1 Backend

- **Framework:** Laravel 11.x (PHP 8.3)
- **API Layer:** Laravel REST API (JSON), versioned (/api/v1)
- **Authentication:** Laravel Sanctum (SPA/API token auth) with role-based guards
- **Authorization:** Spatie laravel-permission for RBAC (roles & granular permissions)
- **Queue & Jobs:** Laravel Queues (Redis driver) for OTP, notifications, document processing, GST-call retries
- **Scheduler:** Laravel Task Scheduling for re-verification reminders, score recalculation, document-expiry checks
- **Search:** Laravel Scout + MySQL full-text (Meilisearch optional upgrade for advanced filters)
- **File Storage:** Laravel Filesystem — local for dev, S3-compatible object storage for production documents
- **Audit Trail:** Spatie laravel-activitylog for immutable audit logging

3.2 Frontend

- **Public website & dashboards:** Blade + Livewire 3 (tight Laravel integration, fast delivery) — OR Vue 3 + Inertia.js if a more app-like SPA feel is preferred
- **Styling:** Tailwind CSS
- **Charts (score breakdown, admin analytics):** Chart.js

3.3 Database & Infrastructure

- **Database:** MySQL 8.x
- **Cache/Queue:** Redis
- **Hosting:** Linux VPS/Cloud (AWS / DigitalOcean / GCP) with Nginx + PHP-FPM
- **Process management:** Supervisor (for queue workers)
- **CI/CD:** GitHub Actions for automated deployment

3.4 Third-Party / External Integrations

- **GST Verification:** via an authorized GSP/ASP provider or licensed data-verification API (subject to commercial terms — see Assumptions, Section 12)
- **SMS/OTP Gateway:** MSG91 / Twilio (or equivalent)

- **Email:** Transactional email via SMTP/Postmark/SES
- **Document storage:** S3-compatible bucket with signed-URL access control

An abstraction/integration layer will be built around GST and other government-data providers so the underlying provider can be swapped without rebuilding the platform, as required in the original specification.

4. System Architecture Overview

BITAINTEL will follow a modular monolith architecture in Laravel — organized into clearly separated domain modules (Business, Verification, Scoring, Search, Enquiry, Notification, Admin) sharing one codebase and database, with a dedicated integration layer isolating all third-party/government API calls.

4.1 High-Level Layers

- **Presentation Layer —** Blade/Livewire (or Vue+Inertia) views for Public site, Vendor, Admin, Verification-team dashboards
- **Application Layer —** Laravel Controllers, Form Requests (validation), Policies (authorization)
- **Domain/Service Layer —** Services & Actions encapsulating business rules (scoring engine, verification workflow, badge issuance)
- **Integration Layer —** Adapter classes per external provider (GST/GSP-ASP, SMS, Email, future MCA/PAN/MSME/sanctions APIs) behind a common interface
- **Data Layer —** Eloquent models/migrations, MySQL, Redis cache
- **Background Processing —** Queued jobs for OTP, notifications, GST verification calls, score recalculation, document-expiry checks

4.2 Why This Approach

- Faster Phase 1 delivery than microservices, while keeping domain boundaries clean for future extraction
- The integration layer directly satisfies the requirement that government/API providers must be replaceable without a platform rebuild
- Structured Eloquent models (not free-text fields) keep the database AI-ready for Phase 2 (document intelligence, risk scoring, recommendations)

5. Functional Requirements by Module

5.1 Public Website

- Home/landing page — introduction, verification methodology, score explanation, verified-business highlights, CTAs (Register / Search / Login)

5.2 Registration & Authentication

- **Stage 1 (Account):** name, email, mobile, password, OTP/email verification, user-type selection

- **Stage 2 (Business basics):** legal name, trade name, company type, registration no., GSTIN, PAN, CIN/LLPIN, addresses, website, year established, employee/turnover range, industry, category
- Secure login with role-aware redirect (Vendor/ Admin / Verification team)

5.3 Business Information Modules (A–F)

- **Module A — Company Information:** legal identity, registration, addresses, contacts
- **Module B — Management:** directors/partners/promoters/KMP, with consent-based visibility for sensitive personal data
- **Module C — Business Operations:** products, services, manufacturing/trading/distribution capability, certifications
- **Module D — Customer & Supply Information:** categories, regions, distribution network — never public without authorization
- **Module E — Financial Information:** turnover/revenue ranges, banking confirmation, statements where required — separate visibility controls
- **Module F — Compliance:** GST/PAN status, licenses, certifications, litigation declarations where appropriate

5.4 Document Management

- Upload by category (GST certificate, incorporation, partnership/LLP docs, address proof, bank confirmation, manufacturing/regulatory licenses, ISO/product certifications, export-import docs)
- Metadata tracked per document: type, upload date, expiry, status, verifier, verification date, rejection reason, version history

5.5 Government Verification (GST priority)

- GST verification module calling an authorized GSP/ASP/approved provider — GSTIN, legal/trade name, status, taxpayer type, registration date, principal place of business, constitution, cancellation status where accessible
- Pluggable integration layer to add MCA, PAN, MSME/Udyam, IEC, and other registries in future without rebuilding core platform

5.6 Verification Workflow & Status

- 11-step flow: submission → field validation → email/mobile verification → government/API verification → document validation → automated inconsistency checks → verification-team review → approve/reject/clarify → score calculation → badge issuance → profile goes live per visibility rules
- 10 status states: Registered, Information Submitted, Verification in Progress, Action Required, Partially Verified, BITAINTEL Verified, Enhanced Verified, Verification Expired, Suspended, Rejected
- Four-point verification framework: Business Identity, Business Operations, Customer/Supply/Financial Profile, Compliance & Risk

5.7 BITAINTEL Score & Badge

- Weighted 0–100 score: Identity 20%, Government Verification 20%, Operations 15%, Documents/Certifications 15%, Financial/Stability 15%, Compliance 10%, Profile Completeness 5%
- Score categories: Excellent (90-100), Strong (75-89), Good (60-74), Moderate (40-59), Limited/Review (0-39)
- Tiered visibility of score detail for Public / Vendor / Admin, per the original specification
- Verification badge tied to level + date (e.g. 'BITAINTEL Verified — August 2026'), never presented as a credit rating

5.8 Search, Discovery & Comparison

- Search by company name, brand, GSTIN, industry, product/service, location, business type, verification level, score range, certification, company size, year established
- Filters: verified-only, enhanced-verified, score range, industry, location, manufacturing/export capability, certification, company size, business type, years in operation
- Public profile page: header, overview, verification checkpoints, capabilities, contact/enquiry (per visibility settings)
- Side-by-side comparison of 2–4 businesses, excluding confidential fields

5.9 Enquiry System

- Enquiry status lifecycle: New → Viewed → Responded → In Discussion → Closed/Rejected

5.10 Dashboards

- **Vendor:** profile completion, verification progress, score & improvement tips, documents, enquiries, notifications
- **Admin:** platform-wide KPIs (registrations, pending/verified/rejected/suspended counts, ageing, average score, enquiries, complaints, risk alerts), vendor management, verification queue management
- **Verification Team:** assigned queue, document viewer, API verification results, checklist, remarks, approve/reject/escalate

5.11 Complaints, Notifications, Audit

- Complaint types: incorrect info, fake company, misrepresentation, invalid certificate, badge misuse, incorrect score, conduct issue — full investigation → decision → action → closure workflow
- Notifications via email, in-app, and SMS/OTP for all key lifecycle events
- Immutable audit log for every critical action (user, timestamp, IP/device, action, before/after values) — not editable by normal users

5.12 Roles & Permissions (RBAC)

- **Super Admin:** full system access, including scoring rules, API management, role management
- **Admin:** vendor/user/verification management, reports, complaints (restricted from system-level settings)
- **Verification Team:** assigned queue, document/API review, checklist, recommend level

- **Business/Vendor:** profile, documents, verification submission, enquiries
- **Business User:** search, compare, shortlist, enquiries
- **Auditor/Quality Reviewer (optional):** reviews verification decisions and audit logs, flags errors

6. Data Architecture

Phase 1 database will store structured, normalized data (Eloquent models with defined relationships) rather than free-text fields, keeping the platform AI-ready for Phase 2 capabilities such as document intelligence and automated scoring.

6.1 Core Entities

- User
- Business
- Business Owner/Representative
- Business Address
- Product
- Service
- Document
- Verification
- Verification Checkpoint
- Government Verification
- Score
- Score Parameter
- Enquiry
- Complaint
- Notification
- Certification
- Audit Log
- Role
- Permission
- API Integration

6.2 Design Notes

- Financial and customer/supplier data modules carry separate visibility flags, enforced at the query/policy layer, not just in the UI